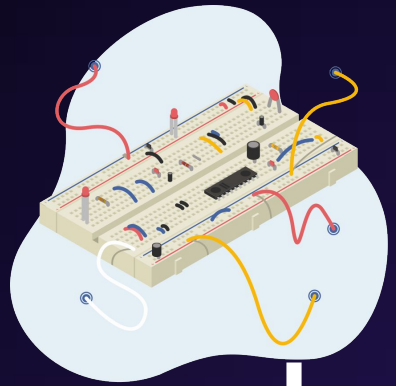
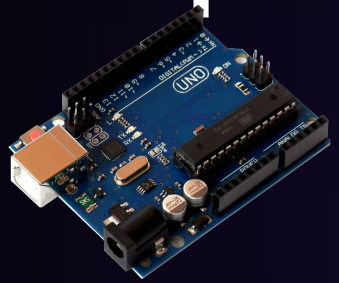
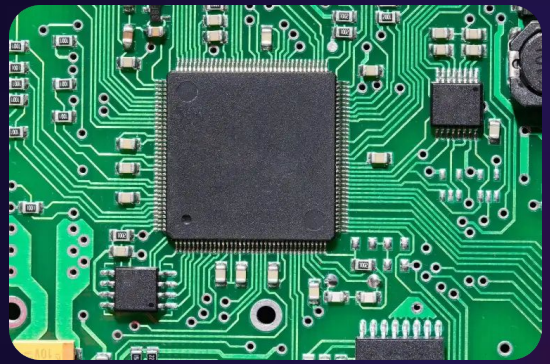
Architecture of electronic and computing machine and computing systems







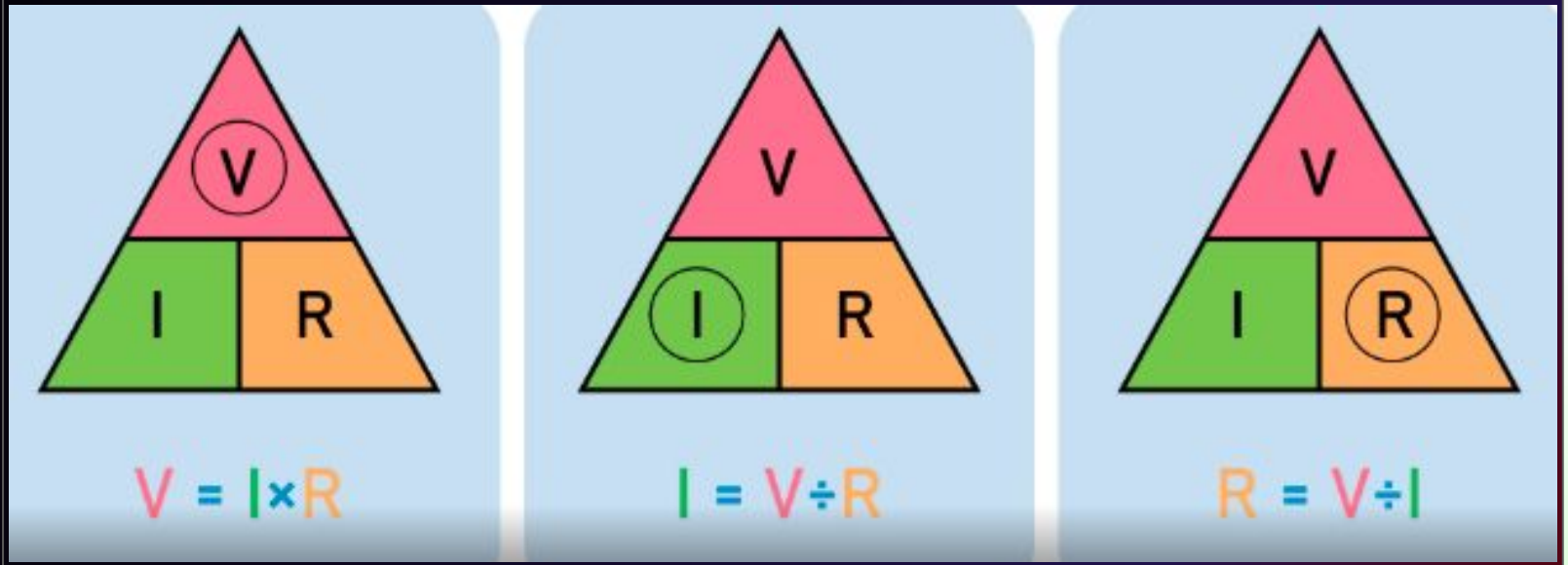
Course Code: S002
Author: Mirlan Nurbekov

ALA-TOO INTERNATIONAL UNIVERSITY
IT & BUSINESS COLLEGE

Agenda

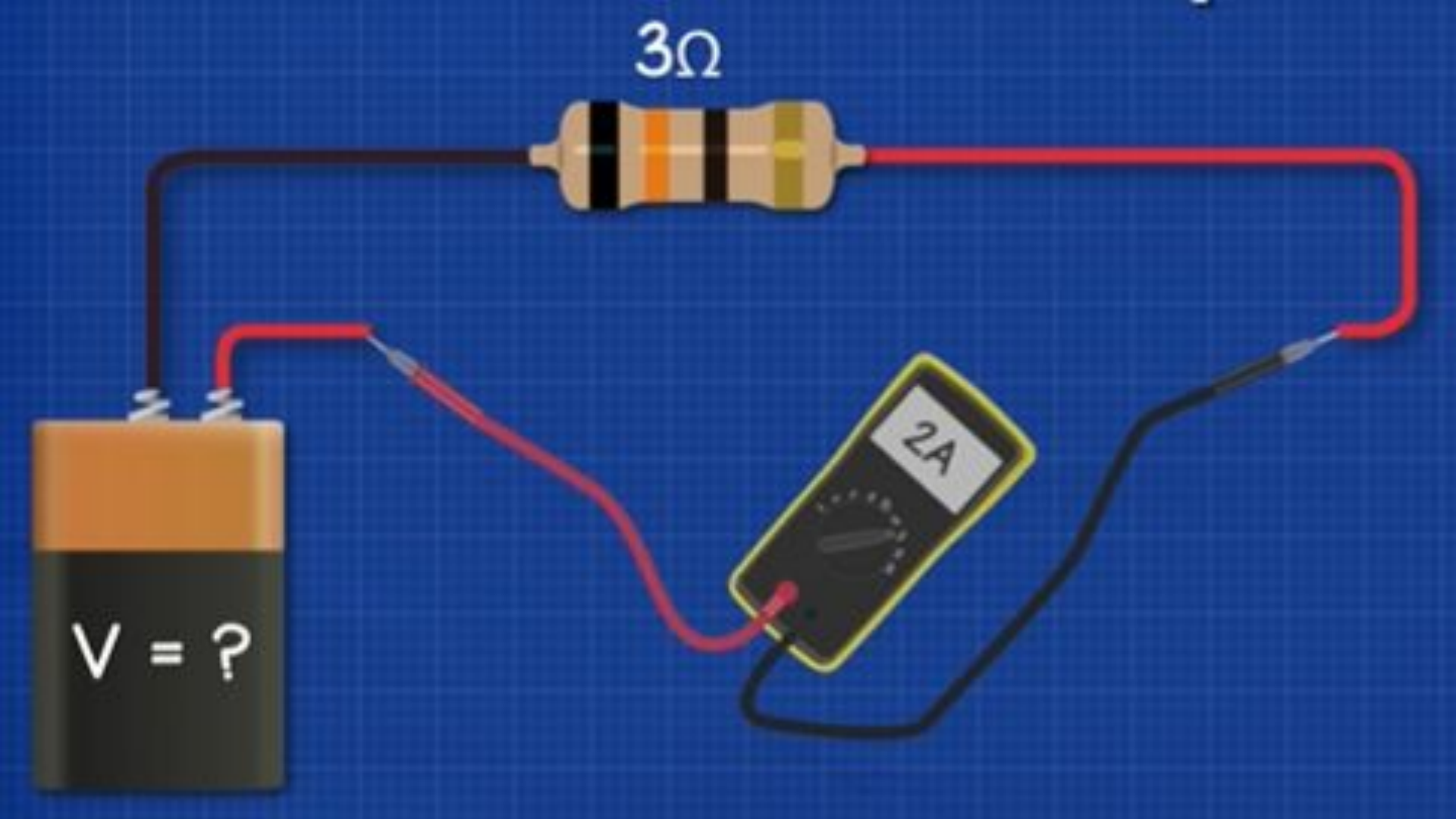
- Digital electronics fundamentals
- Introduction to basic electronic components
- Physics behind jumper cables, buttons, LED and resistors

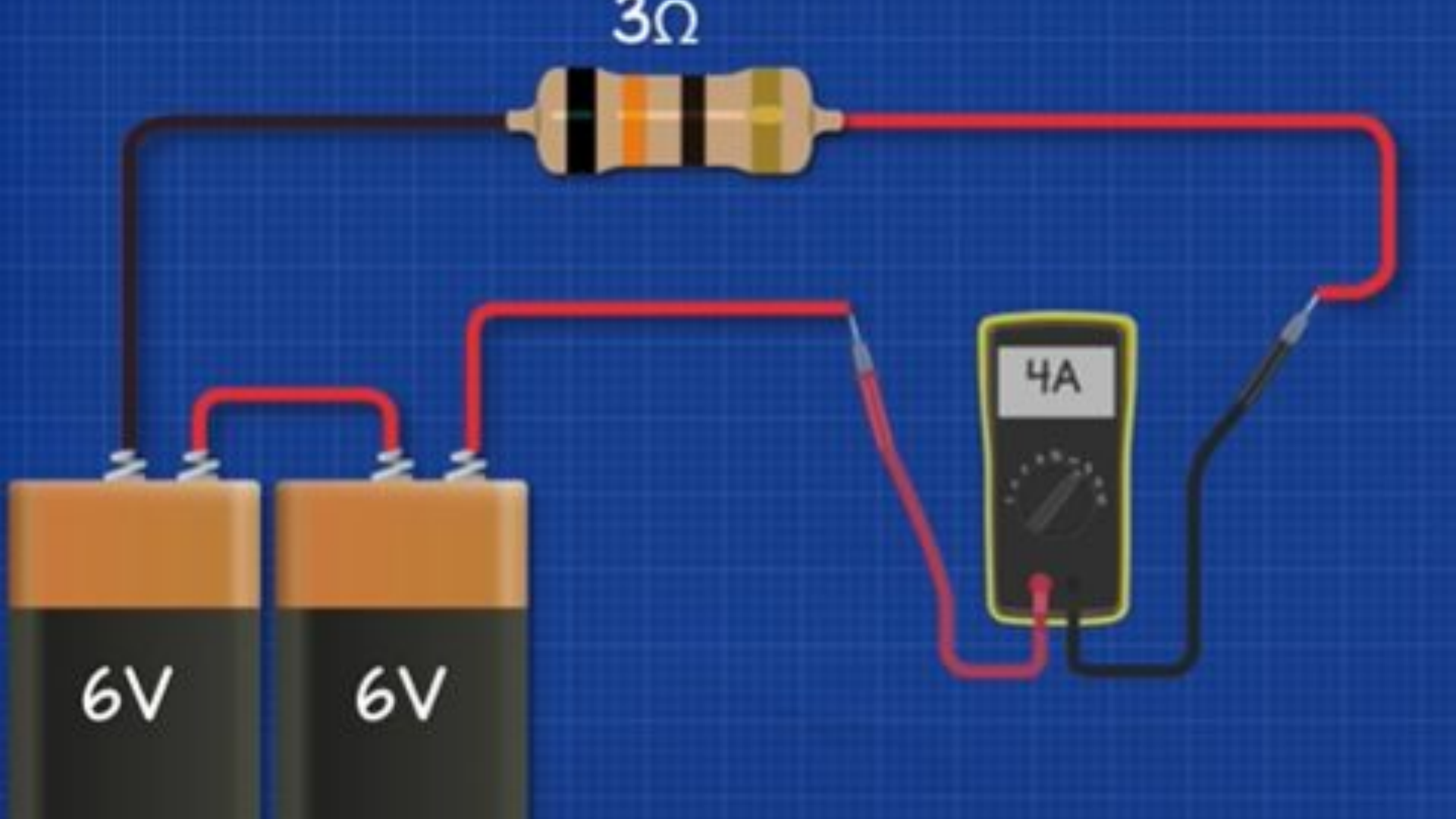
Ohm's Law (V=IR)

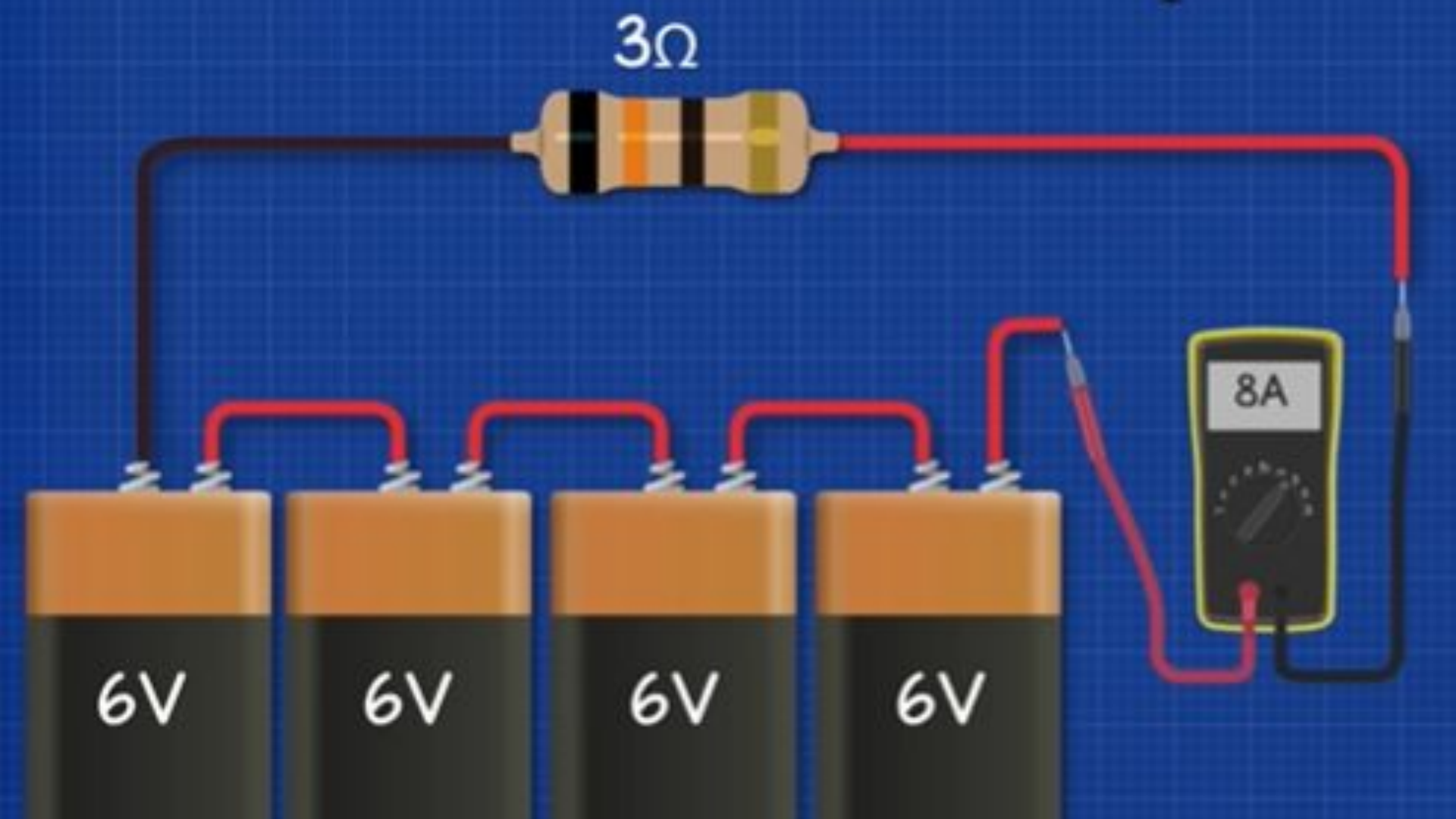


Safety rules in laboratory

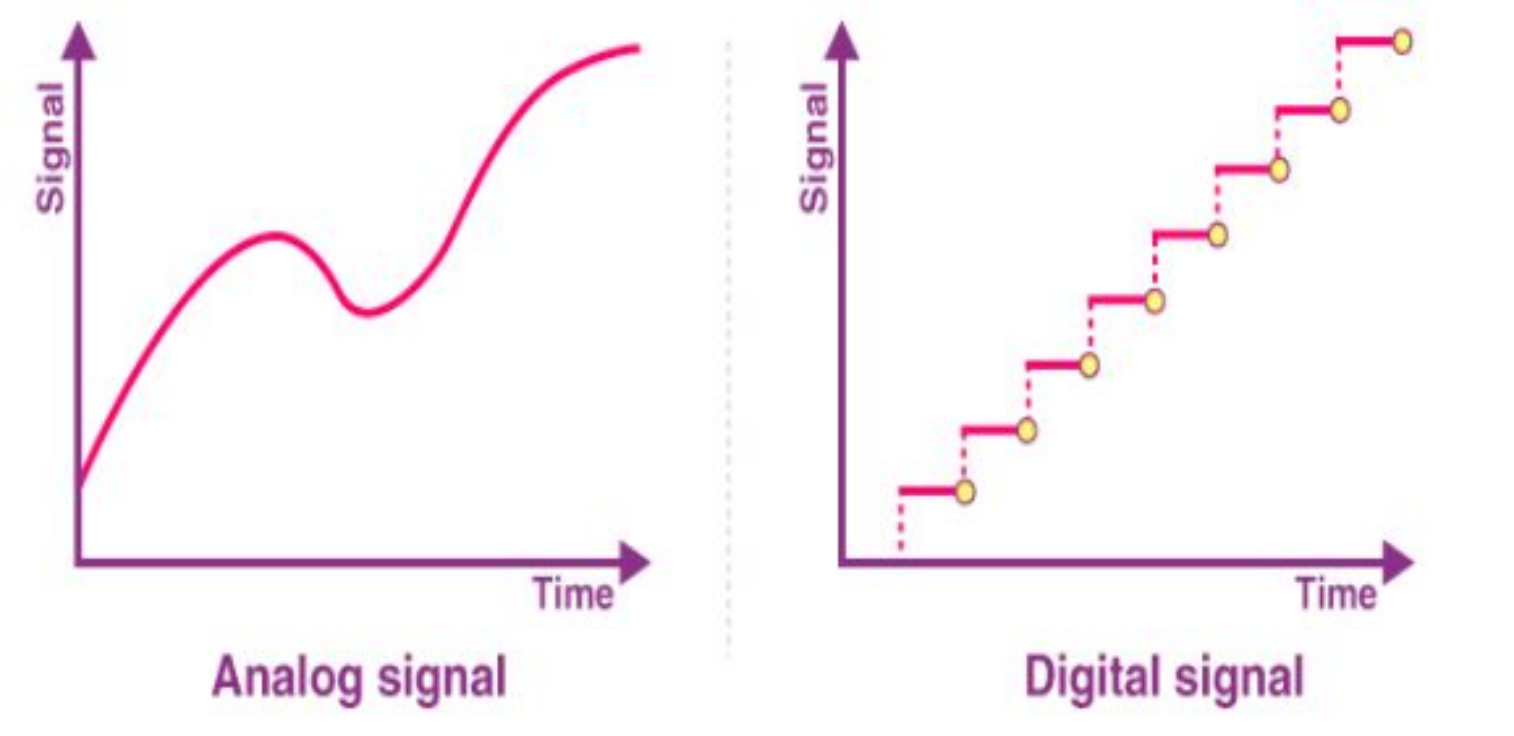


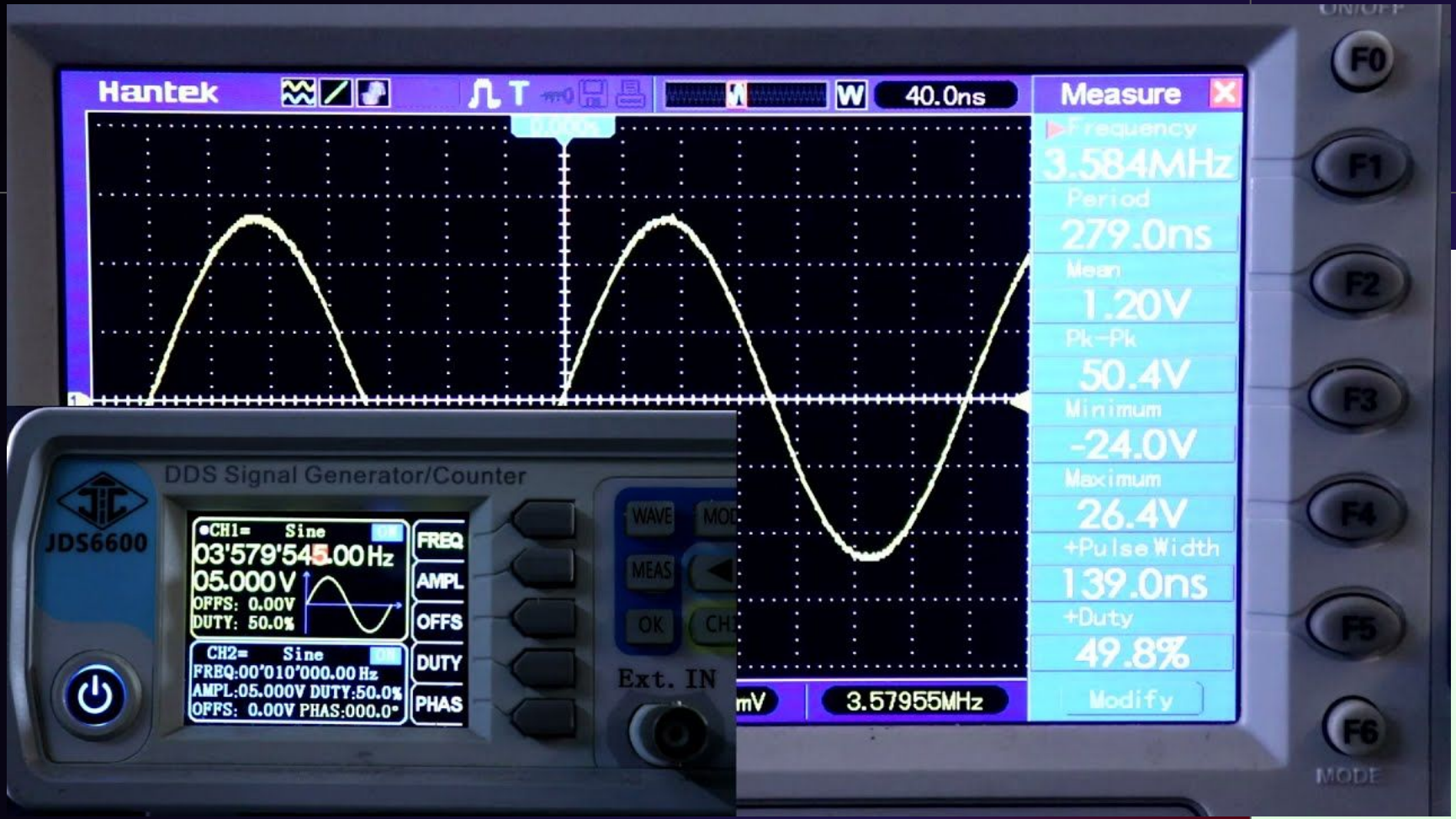

3Ω
2A
V = ?

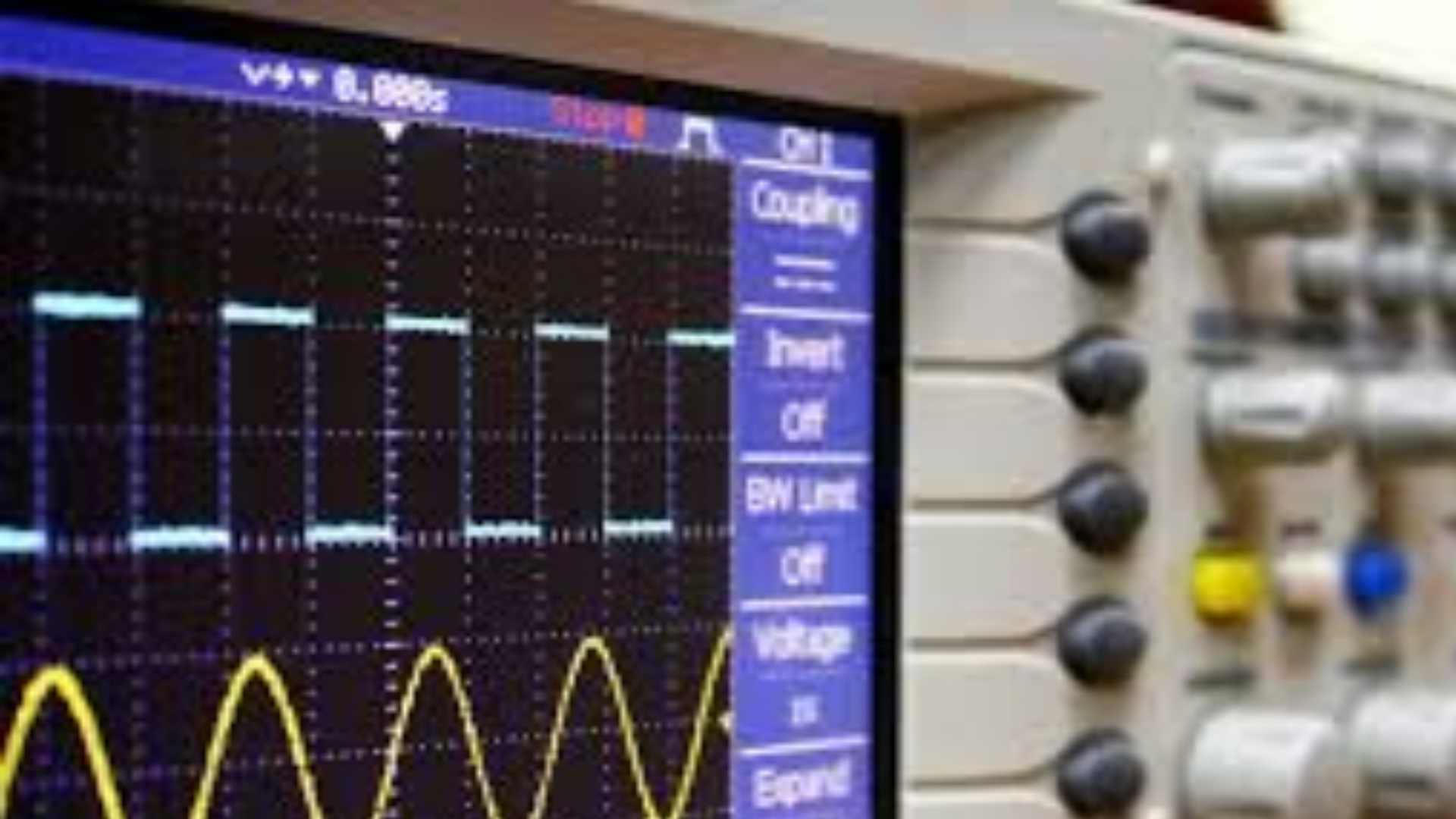

3Ω
4A
6V
6V


3Ω
8A
6V
6V
6V
6V

Analog and Digital signals

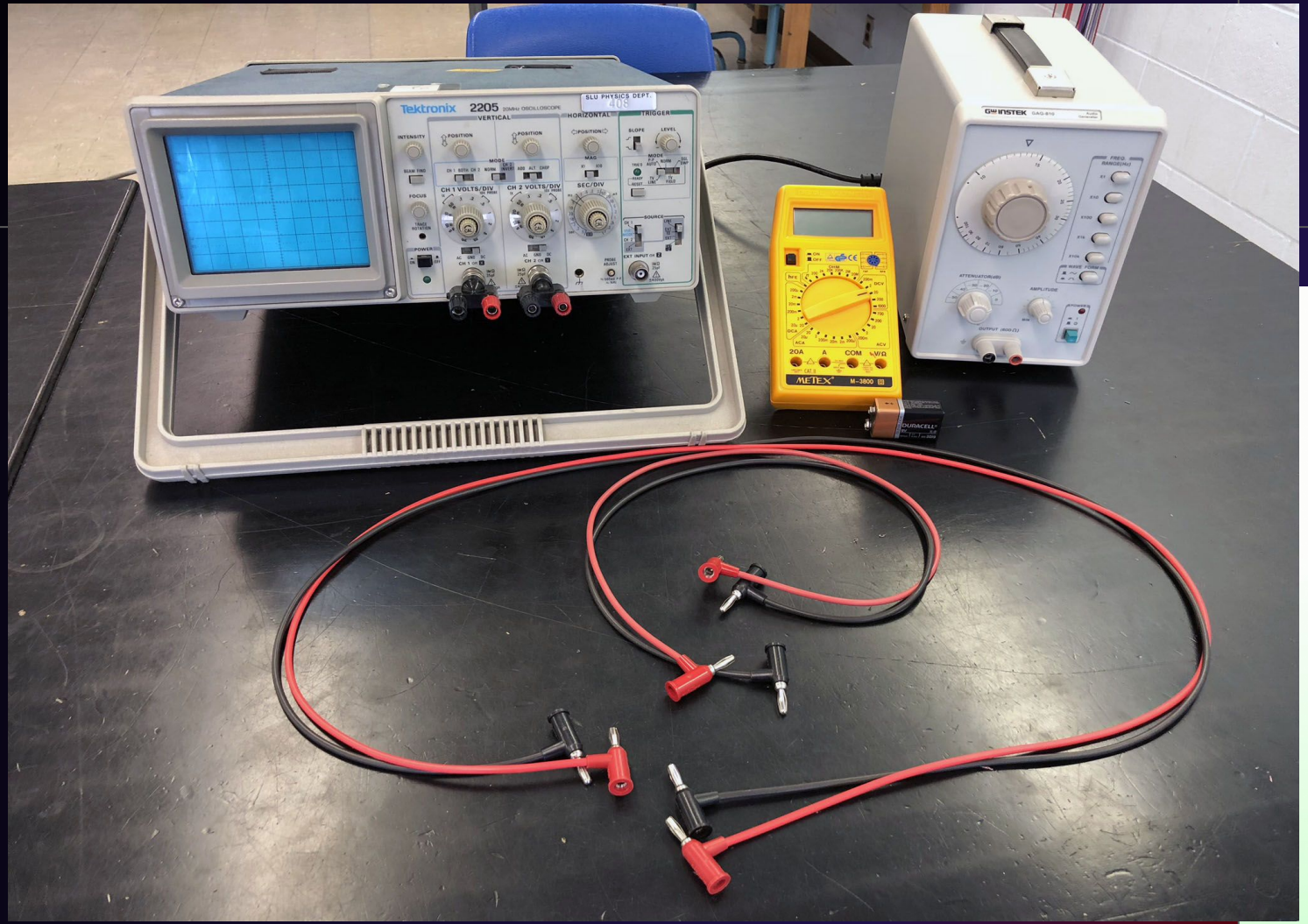



Hantek
40.0ns
Measure
Frequency
3.584MHz
Period
279.0ns
Mean
1.20V
Pk-Pk
50.4V
Minimum
-24.0V
Maximum
26.4V
+Pulse Width
139.0ns
+Duty
49.8%
Modify
3.57955MHz
F0
F1
F2
F3
F4
F5
F6
MODE
DDS Signal Generator/Counter
JDS6600
CH1= Sine
03'579'545.00 Hz
05.000 V
OFFS: 0.00V
DUTY: 50.0%
CH2= Sine
FREQ:00'010'000.00 Hz
AMPL:05.000V DUTY:50.0%
OFFS: 0.00V PHAS:000.0°
FREQ
AMPL
OFFS
DUTY
PHAS
WAVE
MEAS
OK
Ext. IN


Coupling
Invert
Off
BW Limit
Off
Voltage
Expand

Multimeter




Tektronix 2205 20MHz OSCILLOSCOPE
SLU PHYSICS DEPT.
408
VERTICAL
HORIZONTAL
TRIGGER
INTENSITY
FOCUS
POSITION
MODE
CH 1 VOLTS/DIV
CH 2 VOLTS/DIV
SEC/DIV
SLOPE
LEVEL
SOURCE
EXT INPUT
GW INSTEK GAG-810
METEX M-3800
DURACELL

Electronic components



Jumper cables



Jumper cables (Male to Male)



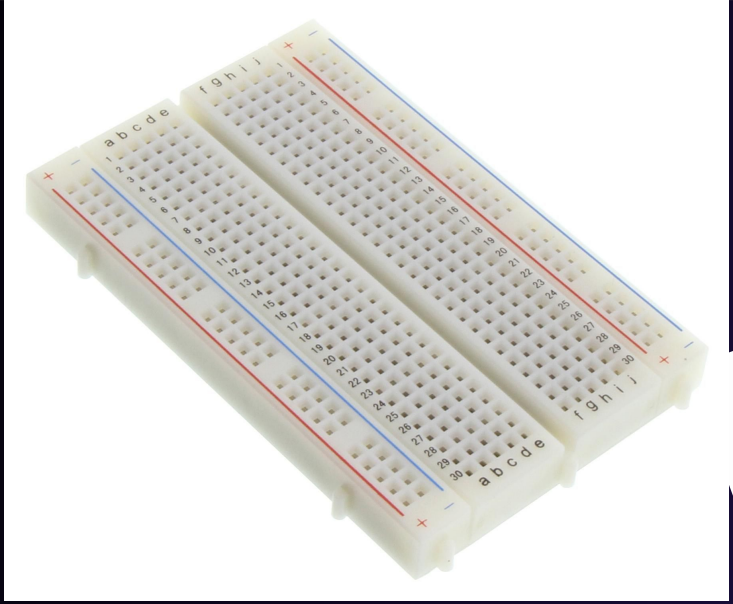
Jumper cables (Male to Female)

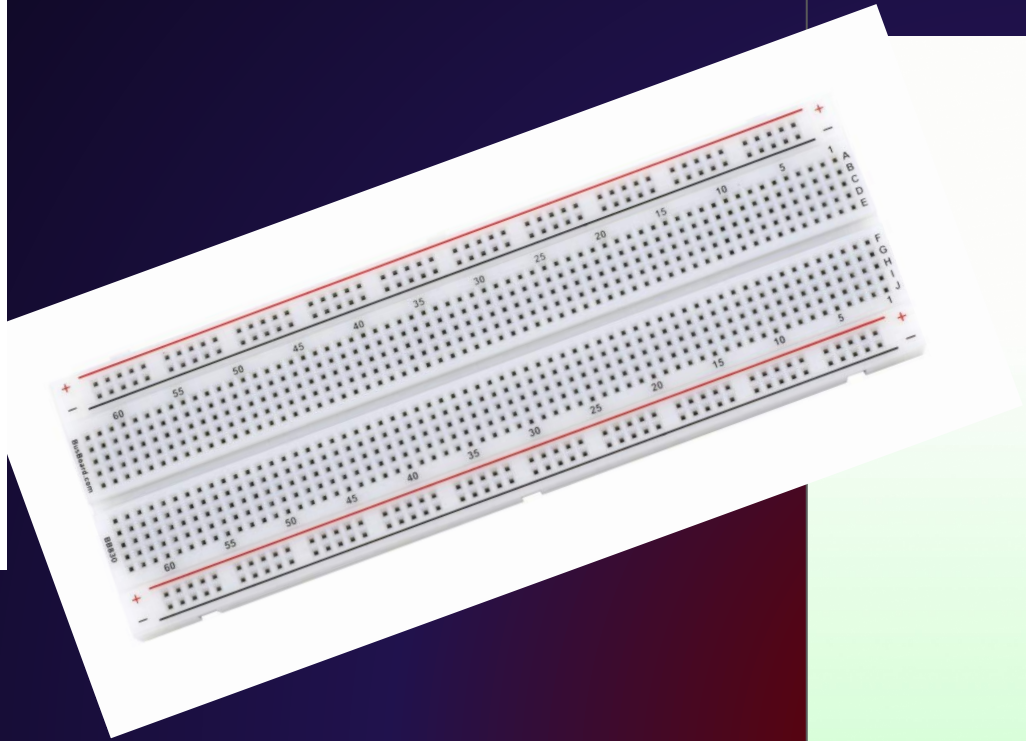


Jumper cables (Female to Female)

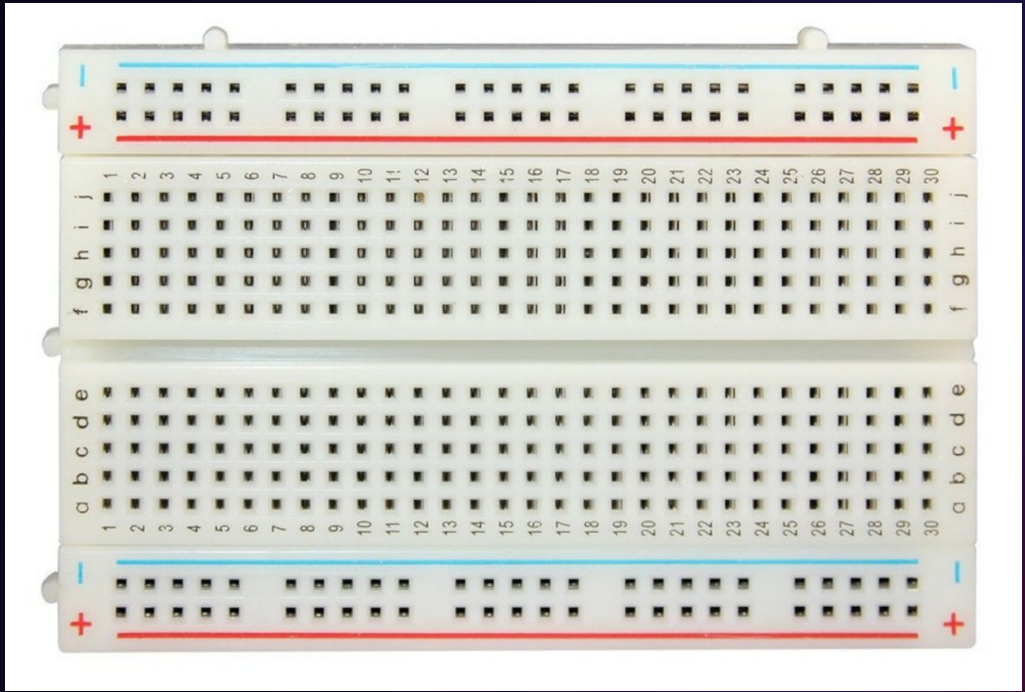


Bread board

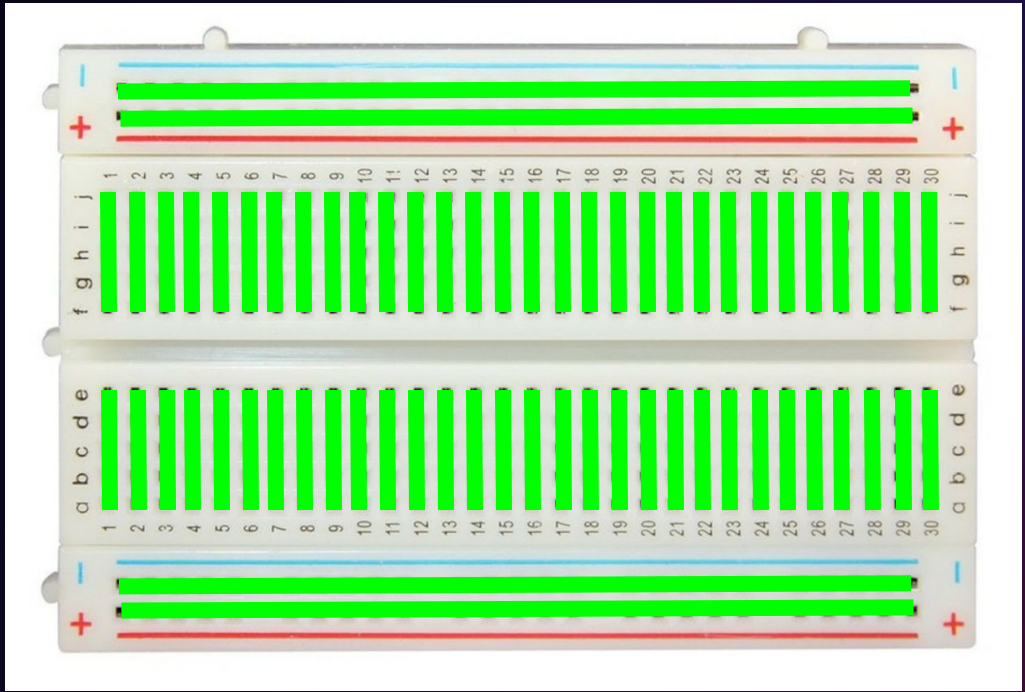




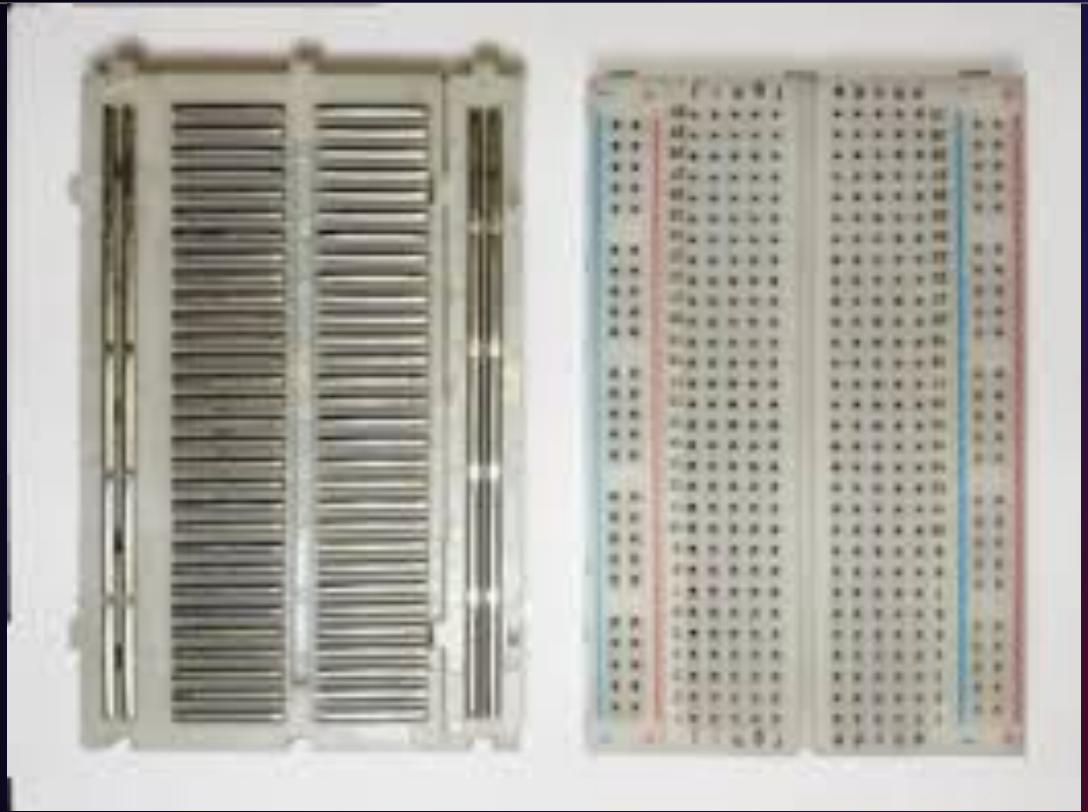
Bread board


a b c d e
f g h i j
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30

Bread board


1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
f g h i j
a b c d e
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30

Bread board

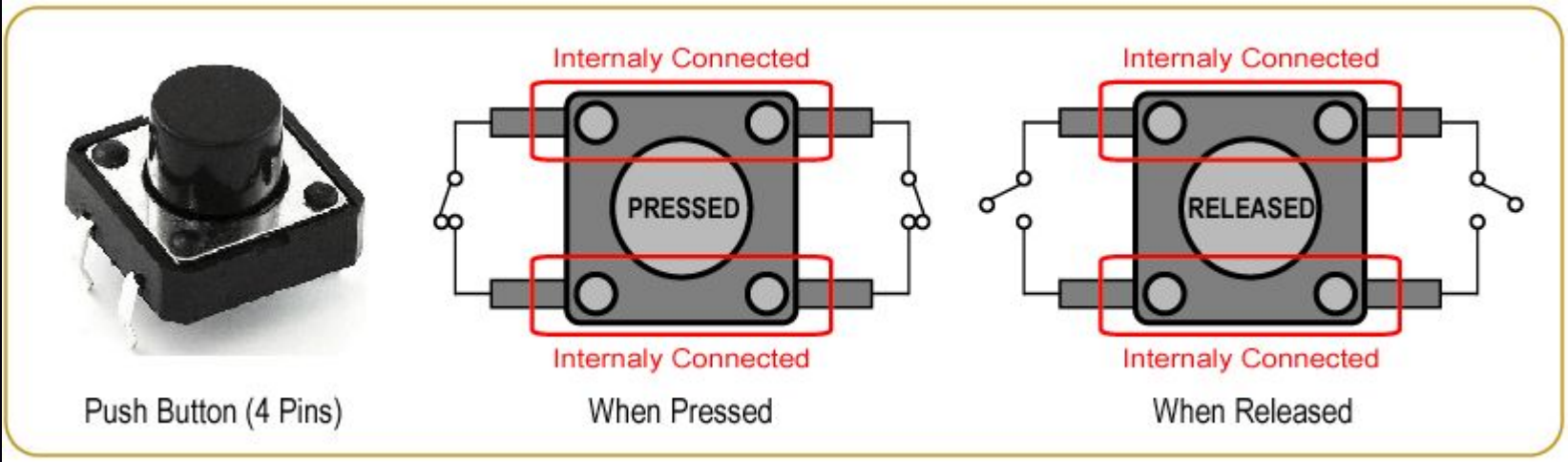


4 pin switch (button)

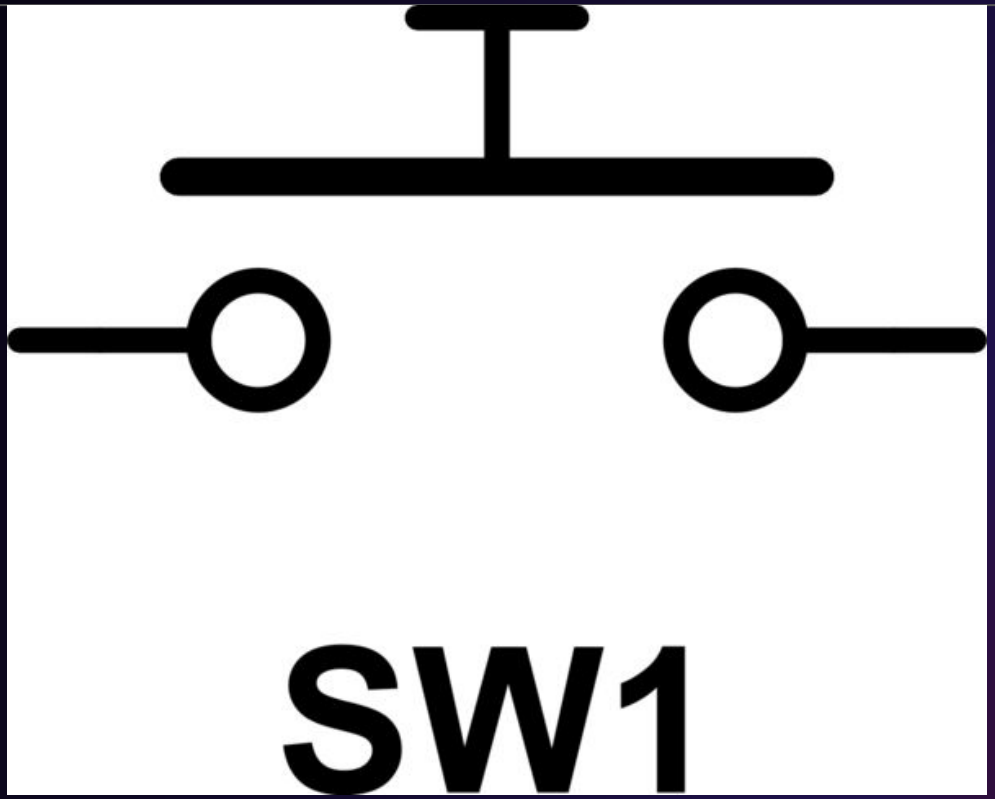




4 pin switch (button)

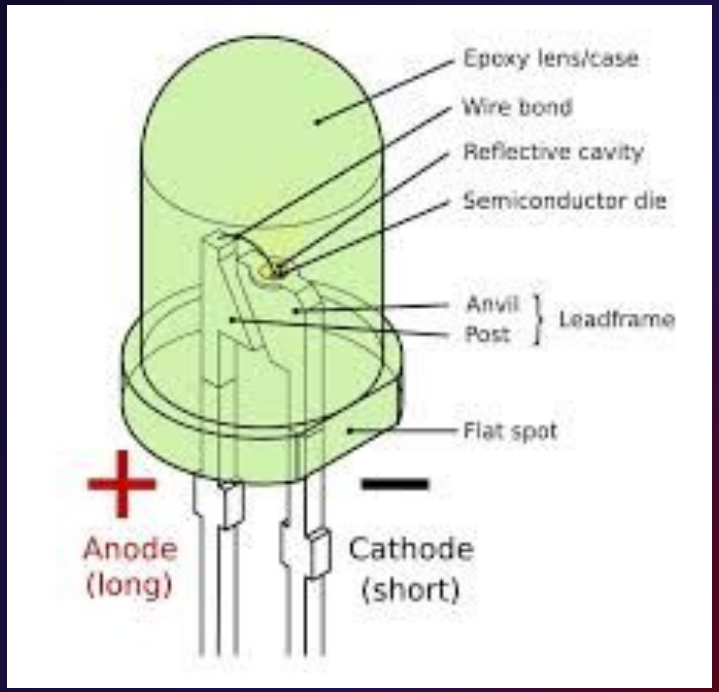


4 pin switch (symbol)



Light Emitting Diode (LED)

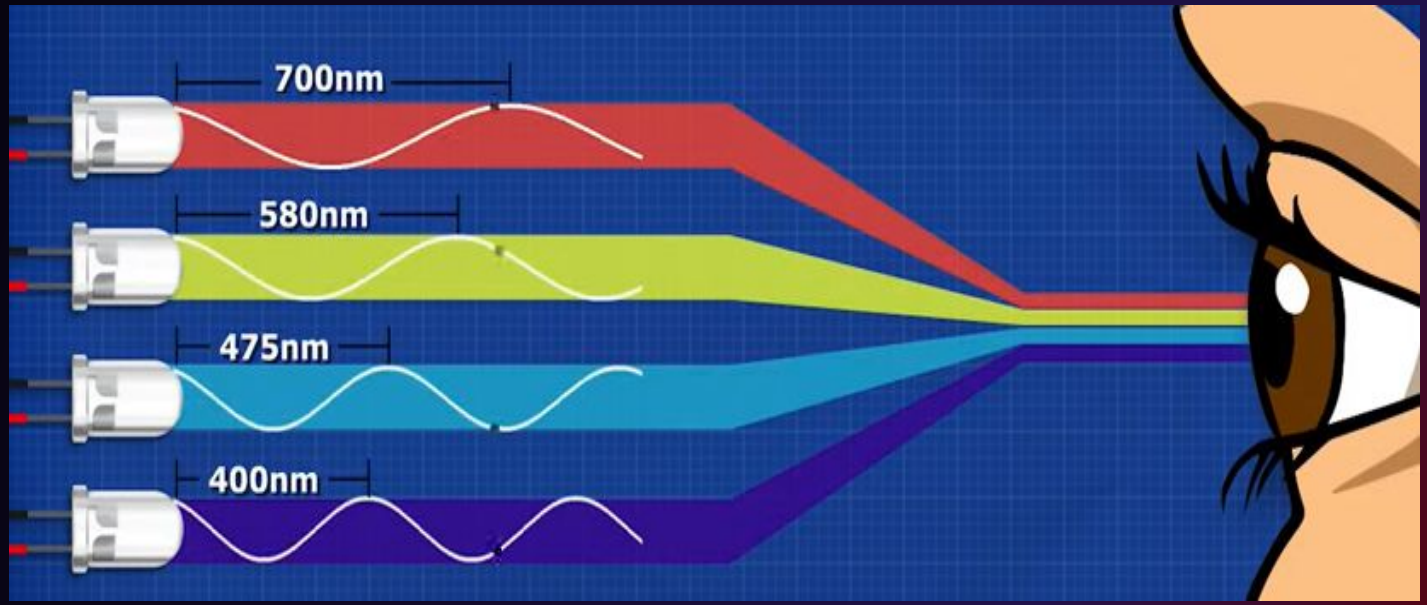




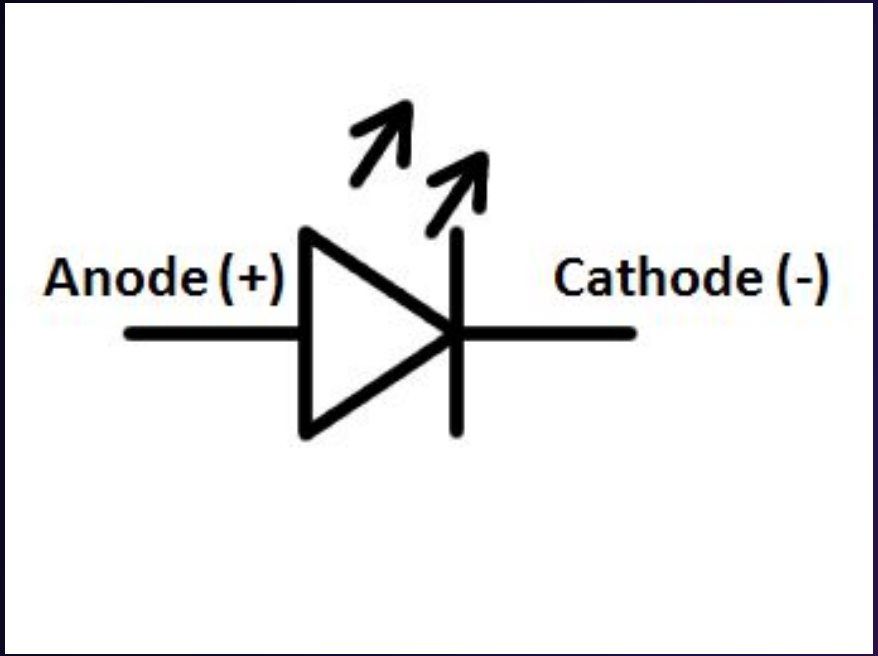
Light Emitting Diode (LED)



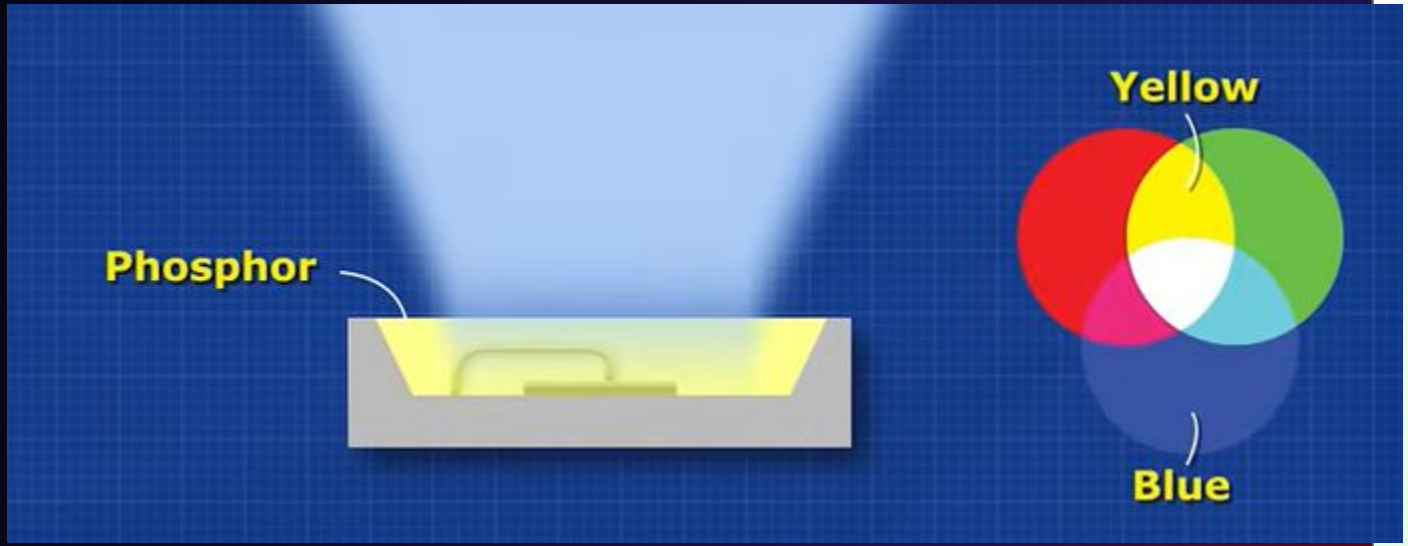
Light Emitting Diode (LED)



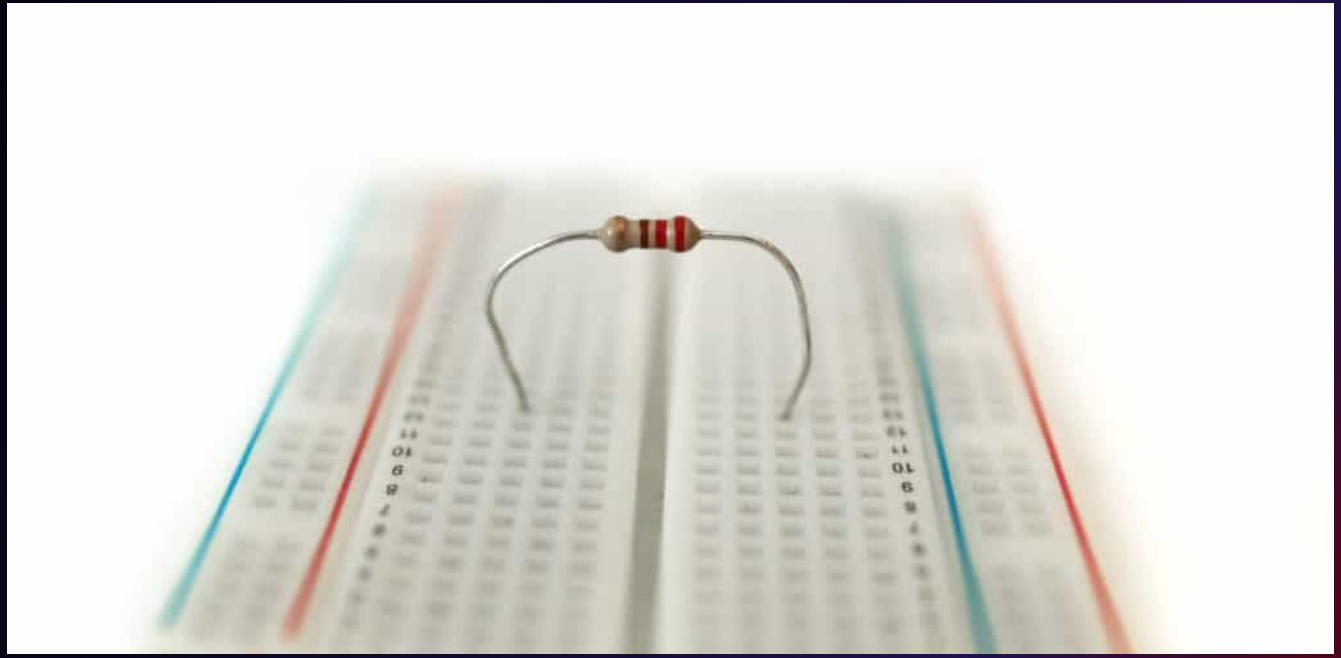
Light Emitting Diode (symbol)



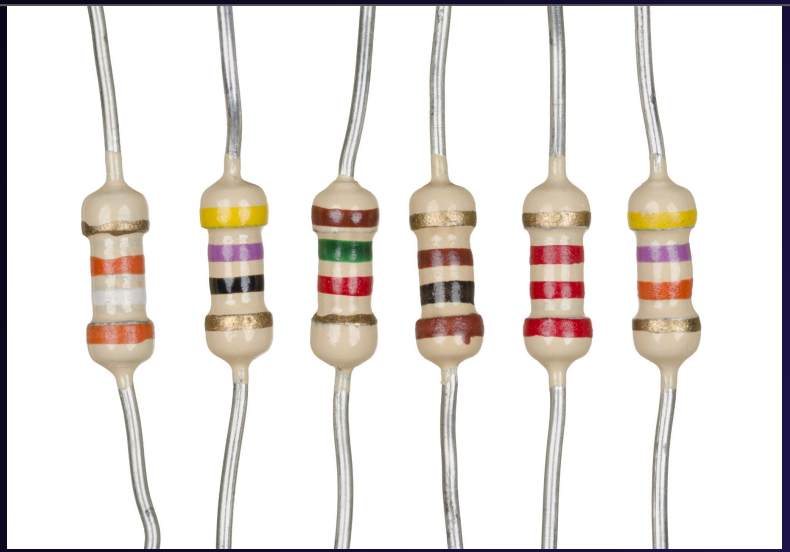
Light Emitting Diode (LED)



Resistors



Resistors



Function: Limits current → protects LEDs & components.
Example: LED needs ~220Ω–330Ω resistor.

Resistors (symbol)



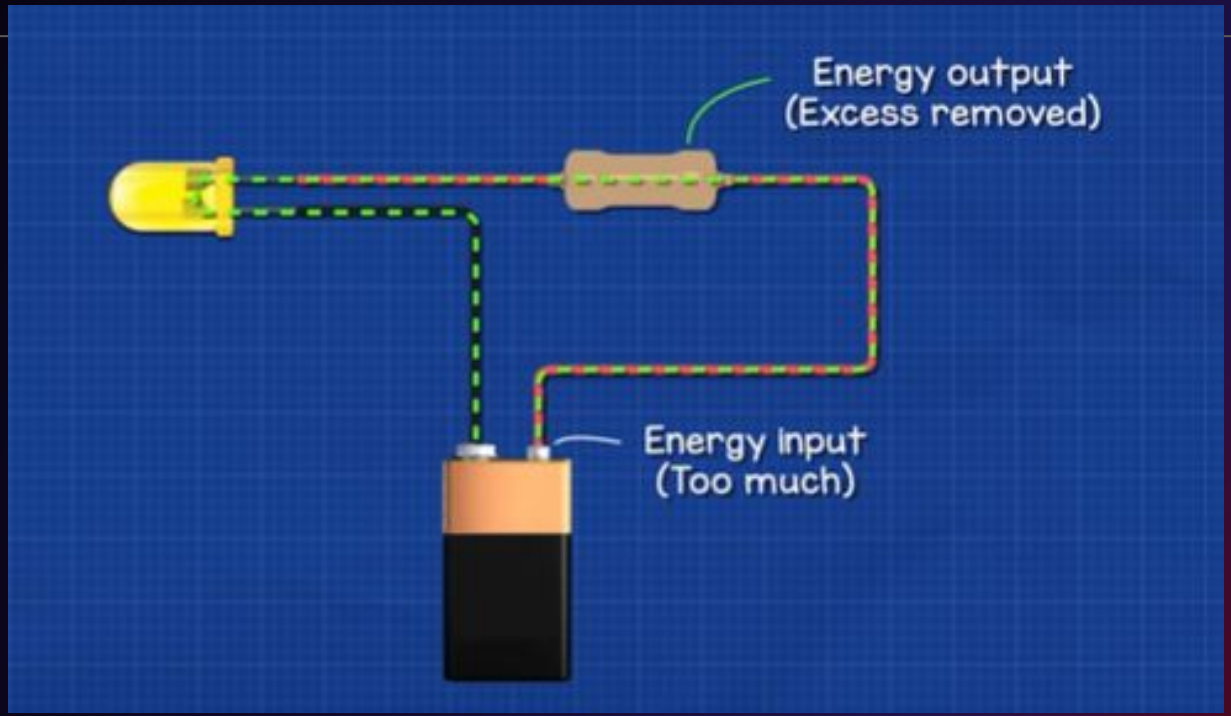
Symbol



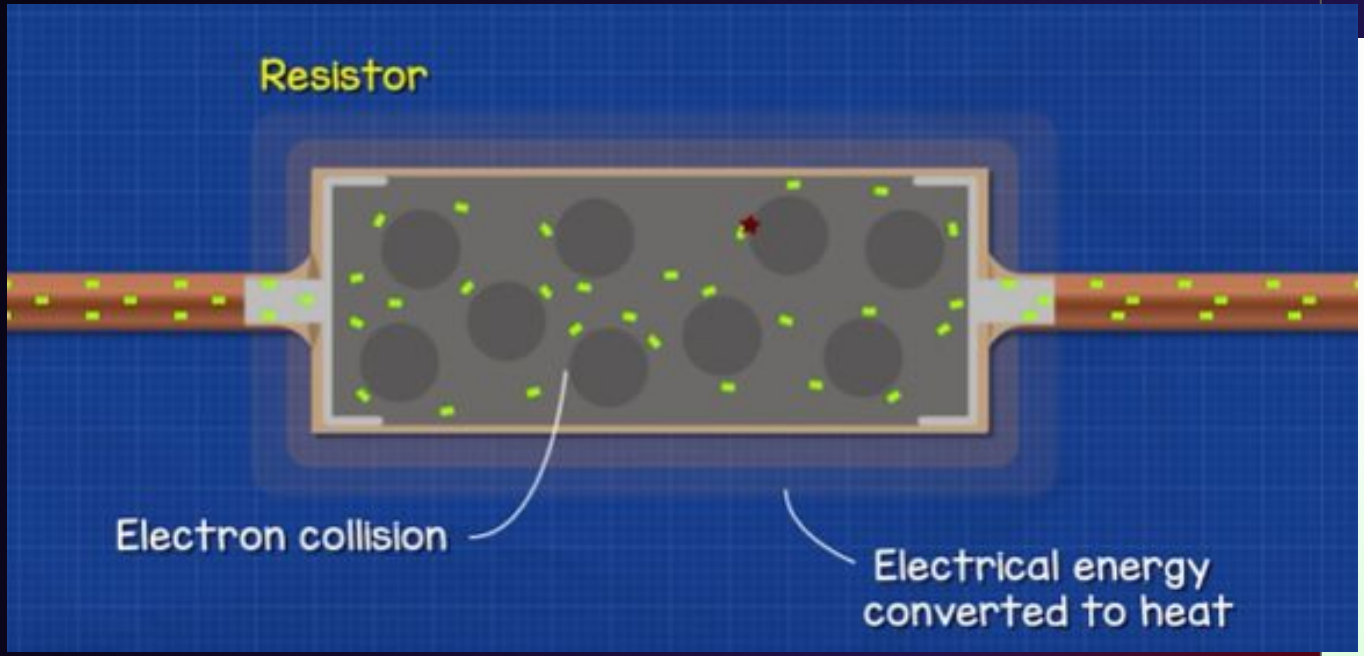
Resistor

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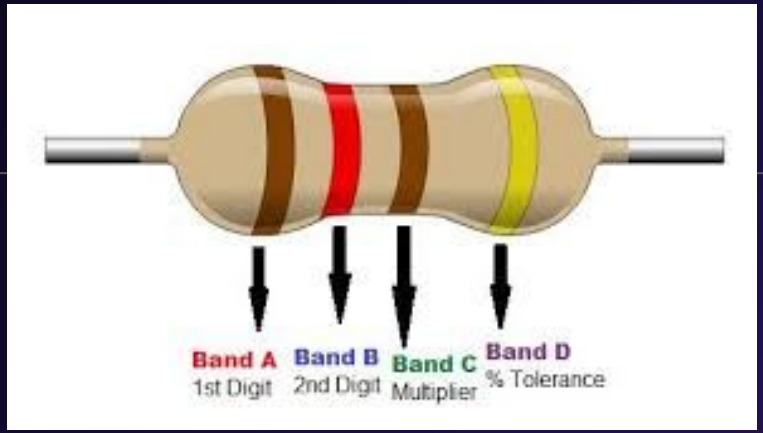
Resistors



Resistors


Resistor
Electron collision
Electrical energy converted to heat

Resistors



Resistor control the flow of electric current in a circuit.
They protect components, divide voltage, and help set the right current levels for electronic devices to work safely and correctly.

100 ohm Resistor

Resistor Color Calculator

Four Band Five Band

Select 1st Band Brown

Select 2nd Band Black

Select 3rd band —

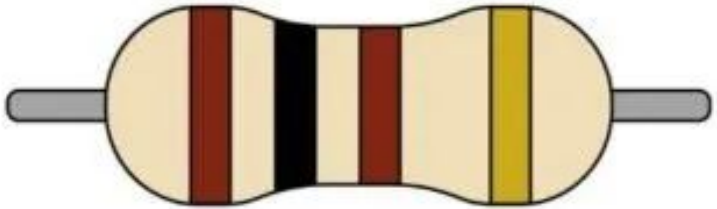
Select Multiplier Brown

Select Tolerance Gold

Calculate Reset

Required resistance is **100 ohms = 100 ohms**.

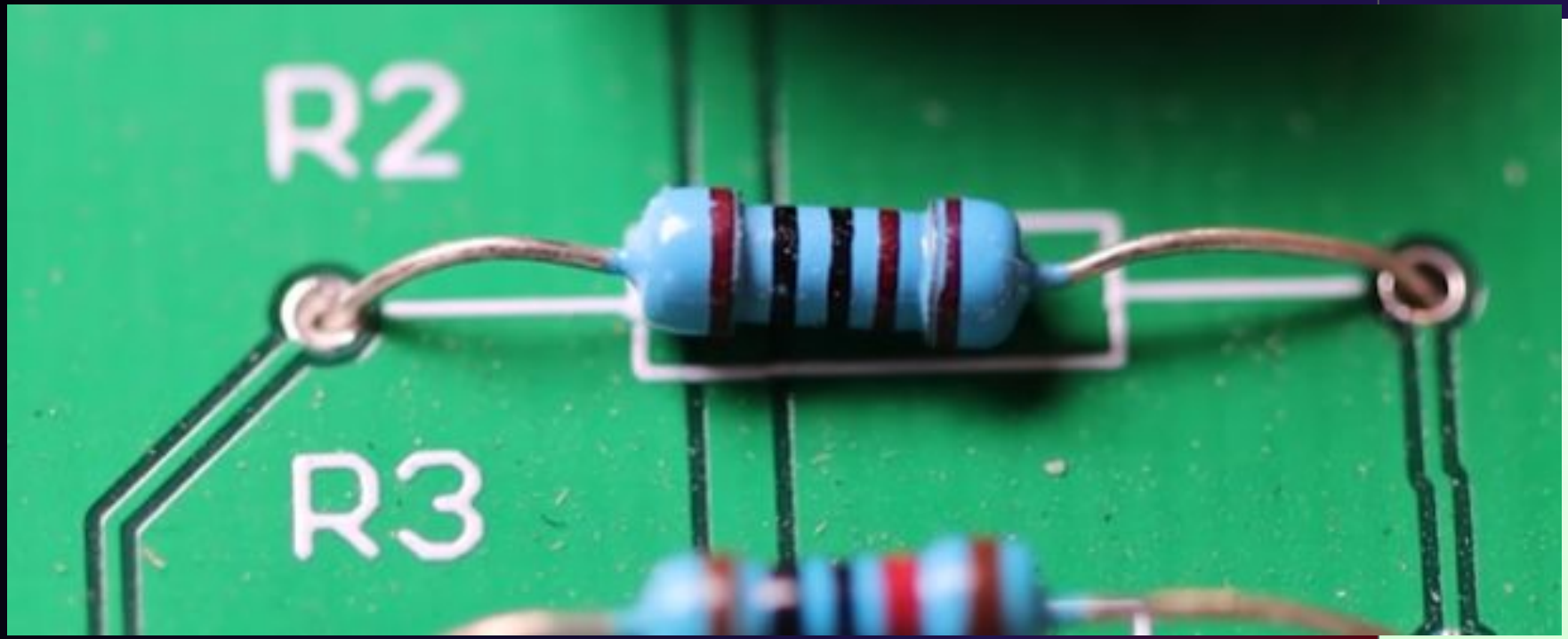
100 Ω





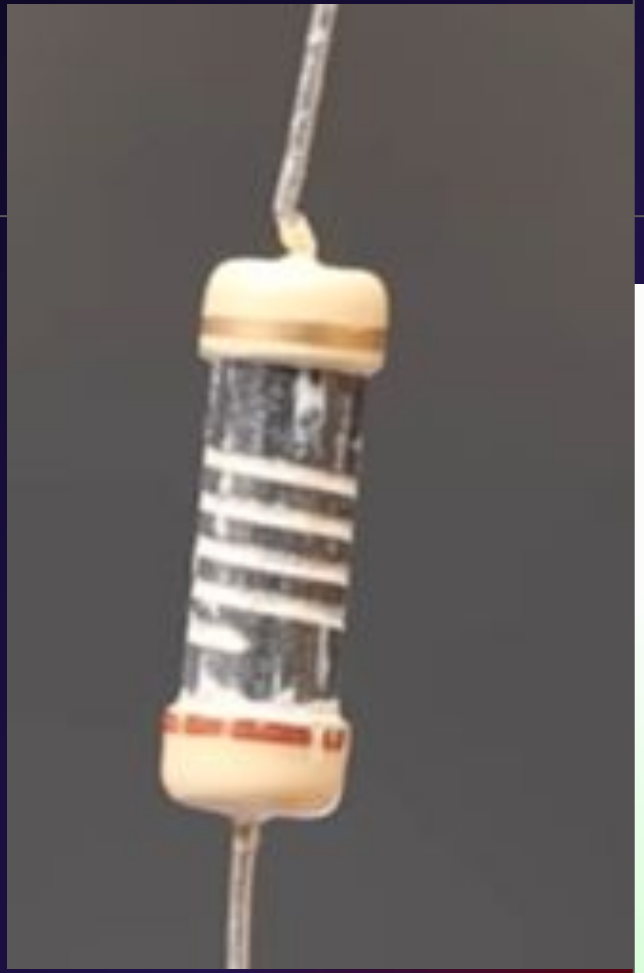


Resistors



Resistors

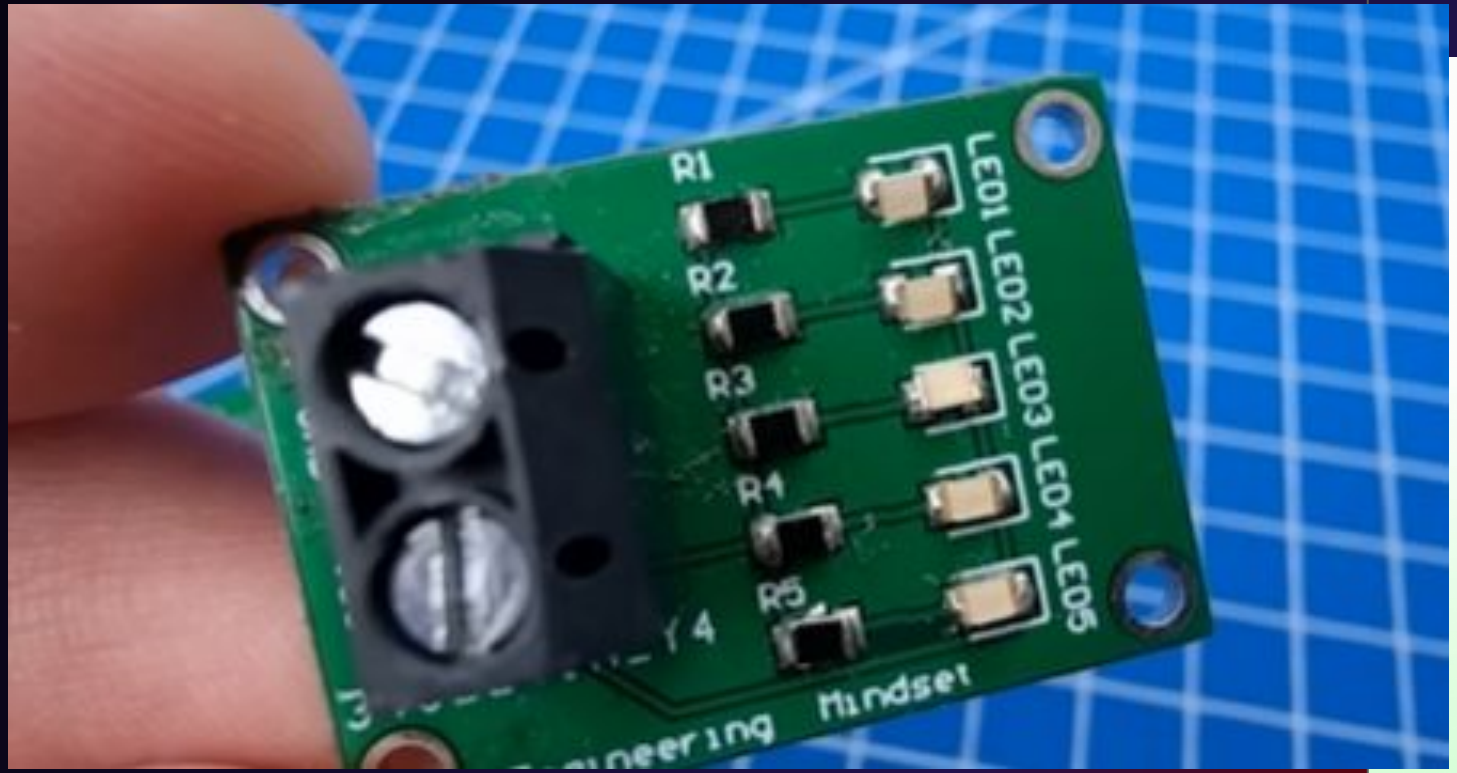




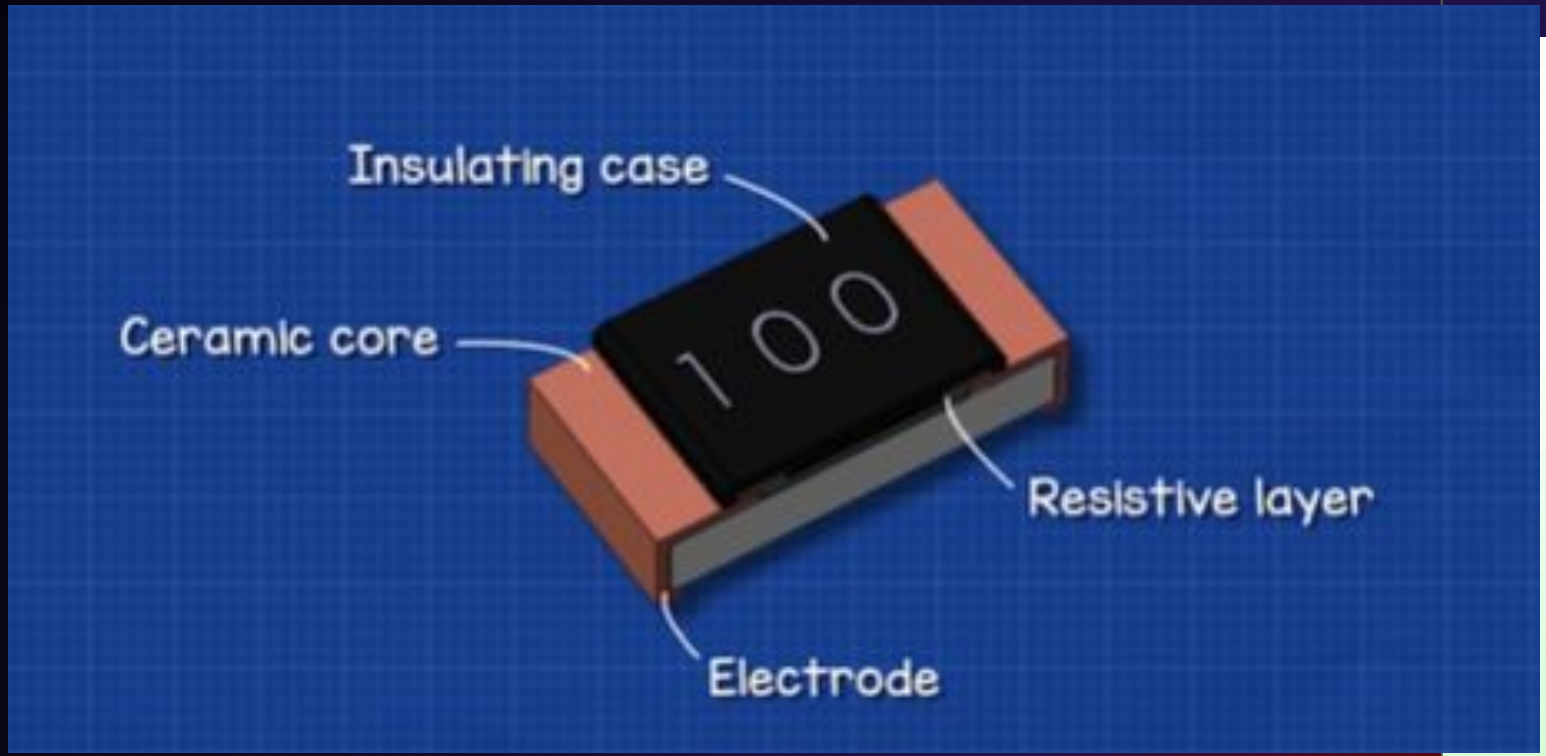
Resistors (SMD)



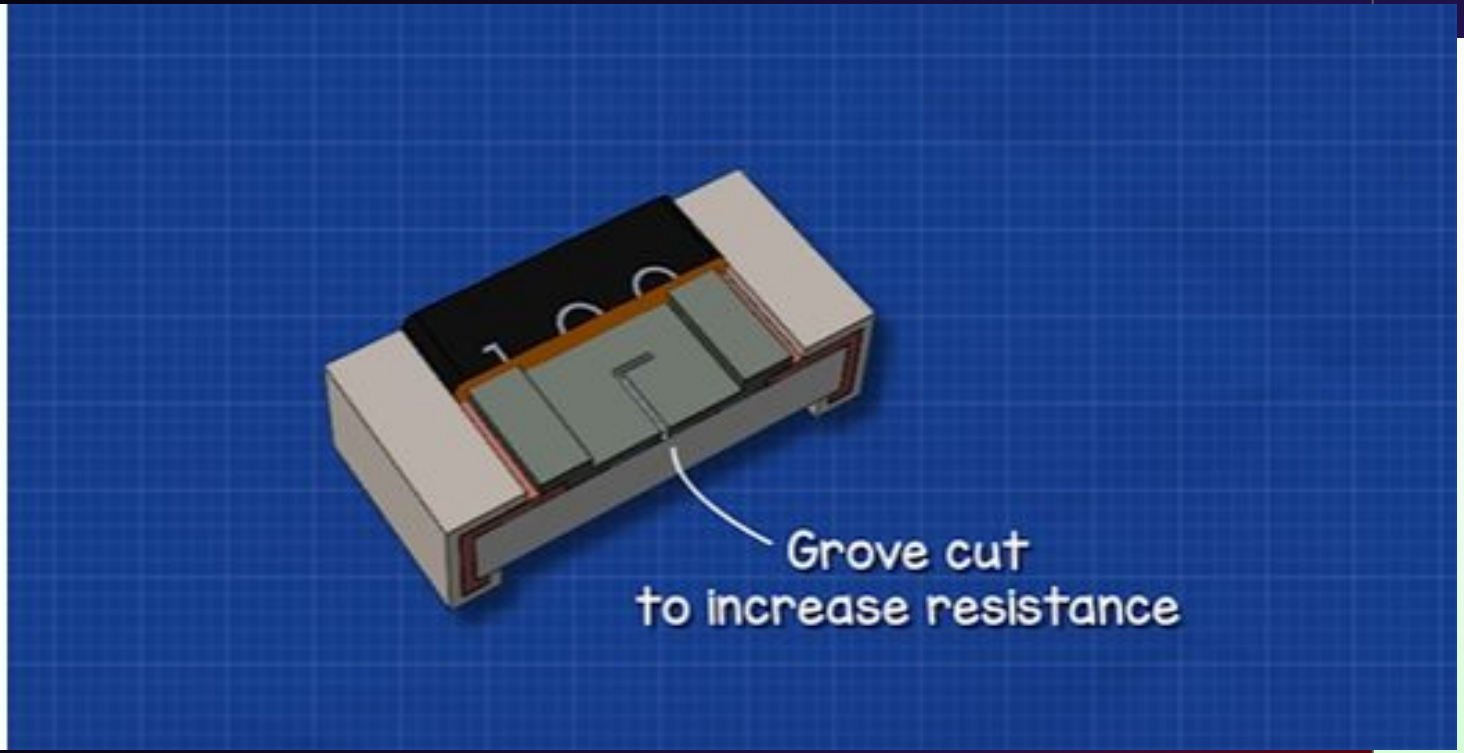
Resistors (SMD)



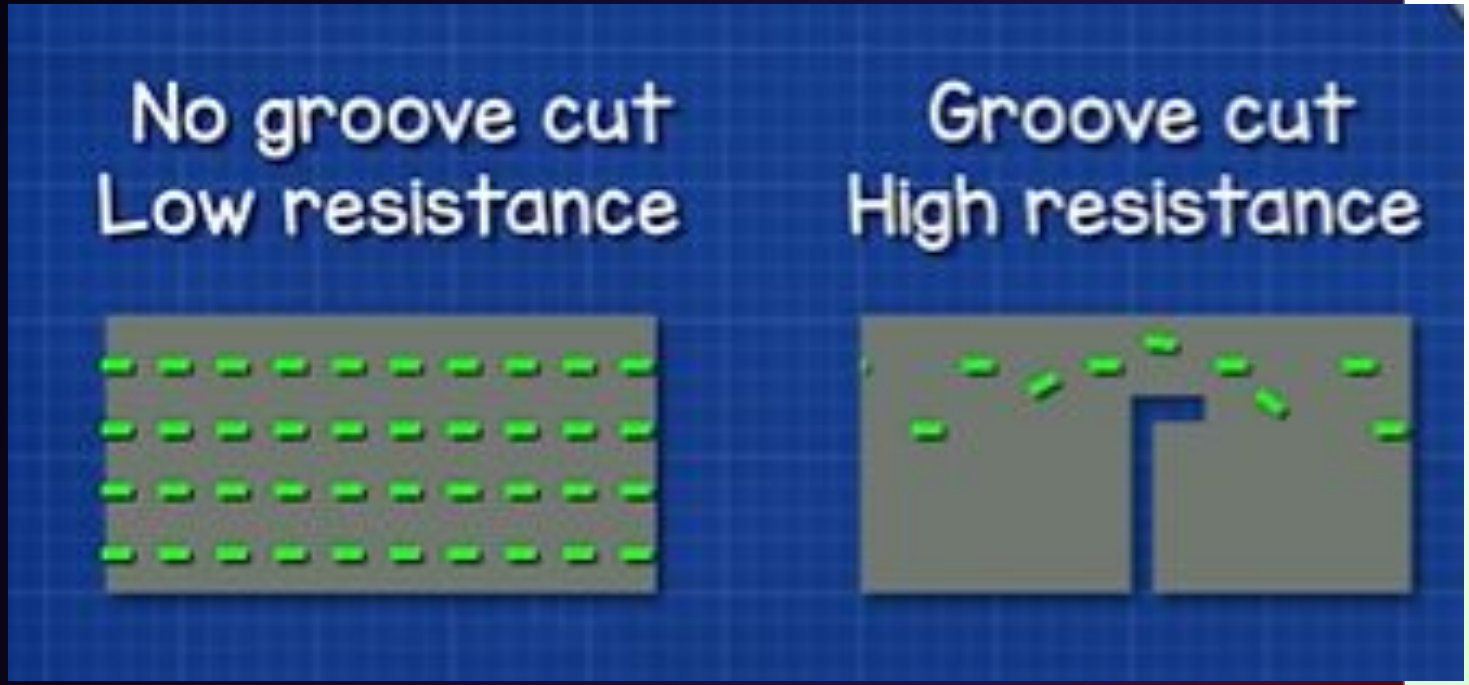
Resistors (SMD)



Resistors (SMD)



Resistors (SMD)



Self test mcq

1. SMD component stands for?

   A) Surface Mounted Device

   B) Small Mechanical Device

   C) Surface Mechanical Device

   D) System Mounted Device

2. Why do we need a resistor?

   A) To increase voltage

   B) To limit current flow

   C) To store energy

   D) To generate electricity

Self test mcq

3. Why are LED frames (lenses) colored?

A) To make LEDs heavier

B) To protect the circuit from super light

C) To indicate color of light emitted

D) To reduce power consumption

Self test mcq

4. How many jumper cable head type combinations are commonly used?

A) 2

B) 3

C) 9

D) 23

Self test mcq

5. What is an SMD component?

A) A component mounted on the PCB surface

B) A component placed inside a battery

C) A component used only for power supplies

D) Type of ice cream

Answers

1. A
2. B
3. C
4. B
5. A

Home work

Review the material covered in today's lesson and prepare a well-organized summary of the key concepts and important points in your notebook. Study the datasheets of the electronic components introduced during the lesson, paying particular attention to their function, pin configuration, electrical characteristics and key specifications. In addition, create an account on Tinkercad or Wokwi (spend some time exploring the platform to become familiar with its interface. Experiment with different components and simple circuits to develop confidence using the simulator before our practical work.